

PRAKHAR SHRIVASTAVA

Mechanical Engineering Undergraduate | SGSITS Indore (RGPV)
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Final-year Mechanical Engineering student specializing in CAD/CAE, mechanism design, and multibody dynamics simulation, seeking a Graduate Engineer Trainee role in the automotive/engineering sector.

EDUCATION

Degree / Certificate	Institute	Year	CGPA / %
B.E., Mechanical Engineering (RGPV, Bhopal)	SGSITS, Indore	2027 (Exp.)	6.93 (till 6th Sem)
Class XII (CBSE)	Maharishi Vidya Mandir, Jabalpur	2022	90.0%
Class X (CBSE)	Maharishi Vidya Mandir, Jabalpur	2020	89.0%

TECHNICAL SKILLS

- CAD/CAM:** Creo Parametric, SolidWorks, Siemens NX, Solid Edge | **Simulation:** MSC Adams, ANSYS, IPG CarMaker
- Programming:** Python, C | **Other:** GD&T, MS Office

WORK EXPERIENCE

CAD/CAE Intern — Avinya Miltech Pvt. Ltd., Indore

Jun 2026 – Jul 2026

- Redesigned the 4-bar mechanism of a lens polishing machine to eliminate quick-return motion, and modelled a Chemical Mechanical Polishing (CMP) machine's kinematics in MSC Adams, incorporating friction, RPM, and trajectory optimization.
- Derived concave/convex conic disc profiles from depth-of-cut vs. radial-distance data and calculated Material Removal Rate using Preston's equation for production planning.
- Performed modal, random vibration, and thermal analyses on an optomechanical assembly in ANSYS; prepared GD&T drawings in Solid Edge and selected bearings to improve service life.
- Built an interactive HTML-based engineering application to predict machine behavior, run calculations, and visualize tool/disc trajectory paths; developed a predictive algorithm for disc re-flattening.

NPD Intern — Gabriel India Ltd., Dewas

May 2025 – Jul 2025

- Verified and consolidated Bills of Material across 1000+ damper assemblies, improving data accessibility and reducing project turnaround time.
- Conducted DFMEA/PFMEA on suspension components and applied Lean Six Sigma and MOST Analysis to streamline assembly-line workflows.
- Studied twin-tube, mono-tube, and strut damper architectures and their mounting systems for real-world OEM applications, gaining working exposure to SAP.

TECHNICAL PROJECTS

Suspension & Steering System Design — Electric ATV (Team GS Racers)

2024 – 2026

- Designed front double-wishbone and rear H-arm suspension in MSC Adams with full hardpoint definition; ran MBD simulations for camber/toe gain, roll centre, Ackermann %, and steering effort.
- Performed four-post rig, sinusoidal ride, and drop-test analyses to extract ride frequencies and upright load cases; designed jigs/fixtures for arm and steering-link manufacturing.

Transmission System Analysis — Electric ATV (Team GS Racers)

2025 – 2026

- Ran dynamic MBD analysis of the gearbox to verify torque transfer, backlash, and shaft alignment, and simulated differential torque split under varying traction conditions.
- Conducted plunge and articulation angle analysis of half-shafts to validate CV joint safety across suspension travel.

Reaction Wheel Attitude Control System for Nanosatellites

Jan 2026 – Mar 2026

- Modeled a reaction wheel assembly in Creo Parametric, simulated rotational dynamics in MSC Adams, and built a 3-axis Euler-dynamics control architecture validated for structural performance in ANSYS.

Rocker-Bogie Suspension for a Planetary Rover

Jan 2026 – Feb 2026

- Modeled a rocker-bogie rover in SolidWorks and implemented it in MSC Adams on a custom rough-terrain test track to evaluate articulation and stability.

LEADERSHIP & ACHIEVEMENTS

- Multibody Dynamics Head, Team GS Racers (2025–26):** Led the MBD division across suspension, steering, and powertrain analysis; trained juniors on Creo, SolidWorks, and MSC Adams.
- 1st Rank, Validation Event E-BAJA 2026; 7th overall rank, M-BAJA 2024 (also competed in E-BAJA'24/'25); AIR 91488 in JEE Mains 2023; School topper, Classes IX–XII.